$(t_0\ \sigma_0\ Q_0\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[((\text{async/lambda}\ (x \ldots\_1)\ e)\ v \ldots\_1)]\ l)\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma_1\ Q_1\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[(\text{task}\ x_{async})]\ l)\ F \ldots)\ FS_1 \ldots))$

where (pt
[task-ret

$(t_0\ \sigma_0\ Q\ (FS_0 \ldots (\text{stack parent-frame}\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma_1\ Q\ (FS_0 \ldots (\text{stack}\ F \ldots)\ FS_1 \ldots))$

where (async? l), $x_{async} = l$, (some x_{child}) = task-tree-running-child$[\![\sigma_0, x_{async}]\!]$, parent-

$(t_0\ \sigma_0\ Q_0\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ v\ l)\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma_1\ Q_1\ (FS_0 \ldots (\text{stack}\ F \ldots)\ FS_1 \ldots))$

where (async? l), $x_{async} = l$, none = task-tree-run

$(t_0\ \sigma\ Q\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[(\text{await}\ (\text{task}\ x_{async}))]\ l)\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma\ Q\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[v]\ l)\ F \ldots)\ FS_1 \ldots))$

where (done v) = task-statu

$(t_0\ \sigma\ Q\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[(\text{await}\ (\text{task}\ x_{async}))]\ l)\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma\ Q\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[(\text{throw}\ v)]\ l)\ F \ldots)\ FS_1 \ldots))$

where (failed v) = task-statu

$(t_0\ \sigma_0\ Q\ (FS_0 \ldots (\text{stack current-frame}\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma_1\ Q\ (FS_0 \ldots (\text{stack}\ F \ldots)\ FS_1 \ldots))$

where (async? l), v_{obj} = lookup$[\![\sigma_0, x_{async}]\!]$, (pending _ ...) = task-status$[\![v_{obj}]\!]$, current-frame = (frame $E[($

$(t_0\ \sigma_0\ Q_0\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[(\text{throw}\ v_{err})]\ l)\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma_1\ Q_1\ (FS_0 \ldots (\text{stack}\ F \ldots)\ FS_1 \ldots))$

where (async? l), (not in-handler?/swift$[\![E]\!]$), $x_{async} = l$, v_{ol}

$(t_0\ \sigma\ Q\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[(\text{cancelled?})]\ l)\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma\ Q\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[\text{is-cancelled?}]\ l)\ F \ldots)\ FS_1 \ldots))$

where (async? l), $x_{async} = l$, is-cancelled? = (task-ancestor-

$(t_0\ \sigma_0\ Q_0\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[(\text{os/io}\ natural\ v)]\ l)\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma_1\ Q_1\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[(\text{task}\ x_{async})]\ l)\ F \ldots)\ FS_1 \ldots))$

where (ptr x_{async}) = malloc$[\![\sigma_0]\!]$, σ_1 = ext1$[\![\sigma_0, (x_{async}$ new-task$[\![]\!])]\!]$, Q_1 = q-push$[\![Q_0$, (frame (os/resolve (task x_{async}) lib:$\Sigma[\![t_0, natural]\!]$ v)

$(t_0\ \sigma_0\ Q_0\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[(\text{os/resolve}\ (\text{task}\ x_{async})\ t_{resolve}\ v)]\ l)\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma_1\ Q_1\ (FS_0 \ldots (\text{stack}\ F \ldots)\ FS_1 \ldots))$

where (>= t_0 $t_{resolve}$), v_{ol}

$(t_0\ \sigma\ Q\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[(\text{os/resolve}\ v_{task}\ t_{resolve}\ v)]\ l)\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma\ Q\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ E[(\text{os/resolve}\ v_{task}\ t_{resolve}\ v)]\ l)\ F \ldots)\ FS_1 \ldots))$

[o

w

$(t_0\ \sigma\ Q_0\ (FS_{main}\ FS_0 \ldots (\text{stack})\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma\ Q_1\ (FS_{main}\ FS_0 \ldots (\text{stack}\ F_{head})\ FS_1 \ldots))$

where all-busy?$[\![FS_0, \ldots]\!]$, $(F_{head}\ Q_1)$ = q-po

$(t_0\ \sigma_0\ Q\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ e_0\ l)\ F \ldots)\ FS_1 \ldots)) \longrightarrow$
$(t_1\ \sigma_1\ Q\ (FS_0 \ldots (\text{stack}\ (\text{frame}\ e_1\ l)\ F \ldots)\ FS_1 \ldots))$

where (not value?$[\![e]\!]$), $(\sigma_1\ e_1) = \Downarrow$base

$(t_0\ \sigma\ Q\ ((\text{stack}\ (\text{frame}\ E[(\text{block}\ (\text{task}\ x_{async}))]\ l))\ FS_{rest} \ldots)) \longrightarrow$
$(t_1\ \sigma\ Q\ ((\text{stack}\ (\text{frame}\ E[v]\ l))\ FS_{rest} \ldots))$

where (sync? l), (done v) = task-statu